

Machine Learning: The No-Nonsense (But Fun) Guide

Day 11: Random Forest Classification – The "Wisdom of Crowds" Algorithm 🌲 🌲 🌲

1. Story & Intuition

Yesterday we built one Decision Tree. It was good—but it had one major problem.

A single Decision Tree has only one opinion. That opinion can sometimes be biased, overfit the training data, or make incorrect predictions.

Now imagine asking 100 experts the same question.

- Each expert studies slightly different data.
- Each expert focuses on different features.
- Every expert gives an opinion.
- The final decision is based on majority voting.

"A single tree might be wrong. A hundred trees voting together? Probably right."

That is the core idea behind Random Forest.

Instead of relying on one Decision Tree, Random Forest combines the predictions of many trees to produce a more accurate and reliable result.

2. What is Random Forest?

Random Forest is an Ensemble Learning Algorithm that combines multiple Decision Trees.

Instead of making predictions using a single model, it builds hundreds of trees and combines their outputs.

Comparison

Single Decision Tree	Random Forest
One model	Hundreds of trees
High variance	Low variance
Can overfit	Less overfitting
One prediction	Majority voting
Less accurate	More accurate
Easier to interpret	Harder to interpret

Two Sources of Randomness

Random Forest introduces randomness in two different ways.

1. Bagging (Bootstrap Aggregating)

Each tree is trained using a random sample of the training dataset.

2. Random Feature Selection

Each tree can only use a random subset of features while creating splits.

Together, these techniques create diverse trees that make better collective decisions.

3. Working of Random Forest

Step 1 – Create Random Datasets (Bagging)

Original Dataset

[A, B, C, D, E, F, G, H]

Random samples for different trees

Tree 1

[A, B, B, D, E, G, G, H]

Tree 2

[A, C, C, D, F, F, G, H]

Tree 3

[B, B, C, E, E, F, H, H]

...

Since sampling is done with replacement, some records appear multiple times while others may not appear at all.

This increases diversity among trees.

Step 2 – Random Feature Selection

Instead of considering every feature during every split, each tree only uses a random subset of features.

Example

Tree 1

Feature 1

Feature 3

Feature 5

Tree 2

Feature 2

Feature 4

Feature 6

Tree 3

Feature 1

Feature 2

Feature 5

This prevents every tree from becoming identical.

Step 3 – Majority Voting

Each Decision Tree predicts independently.

Example

Tree 1 → Cat

Tree 2 → Cat

Tree 3 → Dog

...

Tree 100 → Cat

Final Votes

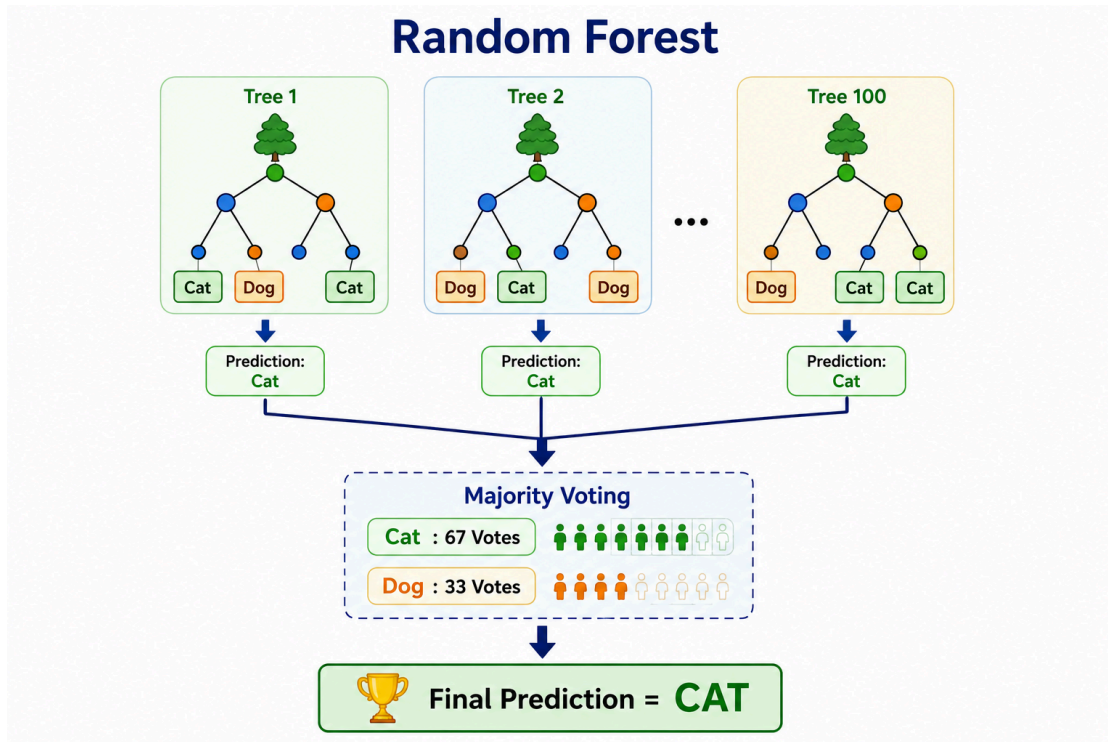
Cat : 67 Votes

Dog : 33 Votes

Final Prediction = Cat

For classification, Random Forest uses Majority Voting.

For regression, it uses the Average of all predictions.



5. Important Hyperparameters

Hyperparameter	Description	Recommended Value
n_estimators	Number of trees	100–500
max_depth	Maximum depth of each tree	10–20 or None
min_samples_split	Minimum samples required for splitting	2–10
min_samples_leaf	Minimum samples in leaf node	1–5

max_features	Features considered at each split	sqrt
bootstrap	Enable bootstrap sampling	True
oob_score	Enable Out-of-Bag validation	True

6. Feature Importance

One of the biggest advantages of Random Forest is that it tells us which features contribute the most to predictions.

Example

Feature	Importance
Petal Length	0.452
Petal Width	0.399
Sepal Length	0.099
Sepal Width	0.050

Higher importance means that feature contributes more to the model's decisions.

7. Out-of-Bag (OOB) Score

Since every tree only uses about 63% of the training data, approximately 37% remains unused.

These unused samples are called Out-of-Bag (OOB) samples.

Instead of creating a separate validation dataset, these OOB samples are used to evaluate the model.

Example

Tree 1 Training Data

[A B B C E E G H]

OOB Samples

[D F]

Each tree has different OOB samples.

The average performance across all trees becomes the OOB Score, providing free model validation.

8. Random Forest vs Decision Tree

Feature	Decision Tree	Random Forest
Accuracy	Good	Better
Overfitting	High	Low
Training Speed	Fast	Slower
Prediction Speed	Fast	Fast
Interpretability	Easy	Hard
Handles Noise	Poor	Excellent
Robustness	Moderate	High

9. When to Use Random Forest

Suitable For

- High prediction accuracy
- Noisy datasets
- Mixed numerical and categorical features
- Feature importance analysis
- General-purpose tabular datasets

Avoid When

- Very small datasets
 - Need complete interpretability
 - Extremely resource-constrained devices
 - Deep learning problems such as images and text
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10. Real-World Applications

- Credit Risk Prediction
 - Medical Diagnosis
 - Fraud Detection
 - Customer Churn Prediction
 - Stock Market Prediction
 - Recommendation Systems
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11. Interview Questions

1. What is Random Forest?
2. What is Bagging?
3. Why is Random Forest called "Random"?
4. How does Random Forest reduce overfitting?
5. What is the OOB Score?
6. What is Feature Importance?
7. Difference between Decision Tree and Random Forest?
8. How does increasing the number of trees affect performance?

- 9. Explain the bias-variance tradeoff in Random Forest.**
 - 10. Compare Random Forest with Gradient Boosting.**
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Happy Learning !!!!